Lecture 22 — Gaussian elimination

Wed Oct 14

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Boyd video (optional): Lecture 31 — solving linear equations · ENGR108 playlist (Boyd)

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Reading: Gaussian elimination (class notes) and Lachniet Ch. 2 (solving systems, RREF). Related objectives: Lectures. Schedule.

Bring paper. Hand work first; software comes Friday. Warm-up → fill-in → board → **Your turn**.

Learning objectives

By the end of class, students should be able to:

1. Perform elementary row operations.
2. Reduce a small augmented matrix to echelon form.
3. Solve a small system using back substitution.
4. Explain why row operations preserve the solution set.

Warm-up

For the system

$$x_1 + 2x_2 = 5$$
$$2x_1 + 3x_2 = 8,$$

write the **augmented matrix** $[A \mid b]$. What does each row mean as an equation?

Fill in

The **augmented matrix** of $Ax = b$ is the matrix $[A \mid b]$ with A on the left and b as an extra _____.

Elementary row operations (EROs) that do not change the solution set:

1. _____ two rows.
2. Multiply a row by a _____ scalar.
3. Add a _____ of one row to another row.

Why they preserve solutions: each ERO is reversible (undo with another ERO), so the new system is _____ to the old one — same solution set.

Gaussian elimination aims for **(row) echelon form**: zeros _____ each pivot, pivots marching right as you go down. Then **back substitution** solves from the _____ equation upward.

Worked in class

W1. One ERO. Starting from

$$\left[\begin{array}{cc|c} 1 & 2 & 5 \\ 2 & 3 & 8 \end{array}\right],$$

replace Row 2 by (Row 2) $-$ 2·(Row 1). Write the new matrix and the matching equation for the new Row 2.

W2. Finish and back-substitute. Continue W1 to echelon form, then solve for x_2 and x_1.

W3. A 3×3 warm-up. Reduce

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 5 \\ 2 & 1 & 0 & 5 \end{array}\right]$$

to echelon form (you choose the EROs). Do **not** fully RREF unless time; stop when back substitution is clear. Solve.

W4. Why not "just swap randomly"? Explain in one sentence why multiplying a row by 0 is **not** an allowed ERO for solving systems.

Your turn

Y1. Augmented matrix for

$$\begin{aligned} x_1 - x_2 &= 1 \\ 3x_1 + 2x_2 &= 12. \end{aligned}$$

Y2. Perform one elimination step: clear the $(2,1)$ entry of

$$\left[\begin{array}{cc|c} 1 & -1 & 1 \\ 3 & 2 & 12 \end{array}\right]$$

by a row replacement. Write the new matrix.

Y3. From the echelon form

$$\left[\begin{array}{cc|c} 1 & -1 & 1 \\ 0 & 5 & 9 \end{array}\right],$$

solve by back substitution.

Y4. True or false, with a one-line reason: if you swap two rows of $[A \mid b]$, the solution set of $Ax = b$ can change.

Answers (Your turn)

Y1.

$$\left[\begin{array}{cc|c} 1 & -1 & 1 \\ 3 & 2 & 12 \end{array}\right].$$

Y2. Row2 $\leftarrow$ Row2 $-$ 3·Row1 gives

$$\left[\begin{array}{cc|c} 1 & -1 & 1 \\ 0 & 5 & 9 \end{array}\right].$$

Y3. From Row 2: $5x_2 = 9$ so $x_2 = 9/5$. From Row 1: $x_1 - x_2 = 1$ so $x_1 = 1 + 9/5 = 14/5$. Solution $x = (14/5,\ 9/5)$.

Y4. **False.** Swapping rows is an elementary row operation; it reorders equations but keeps the same solution set.